

Calibrated, Label-Free Screening for Coordinated Manipulation and Best-Response Evaluation of Exchange Rules in a Vietnam-Calibrated Market Simulator

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Abstract

Prosecuted manipulation on the Vietnamese stock exchanges is mostly carried out by groups of accounts that trade in step with each other, yet regulators have no labelled cases to train a detector on, and the order-level data needed to test one is not public. We study which regulatory levers work against such rings in an agent-based limit order book calibrated to published rules and to public price data. We propose a coincidence test for coordinated accounts with an exact circular-shift null and no labels; netting buys against sells makes market-making desks neutral. Its false-alarm rate is at the nominal level in clean markets, it finds a ring among the five top-ranked of about 120 accounts in 85–90% of ring-active windows, but only 52% when honest groups that act in step are present, and its sensitivity falls as accounts grow; a model trained with labels does better everywhere, so the test matters only where labels do not exist. To judge policies we let the ring re-optimize against each rule, across two families of six simulated markets (index level and stock level volatility) that all match the real moments, and compare with what can be measured on public data. Across these markets the daily price band does not change the ring’s gain, pricing off-book block trades at a trailing mean removes most of it (median 11% and 20% of its value at a one-day mean in the two families, but not everywhere), and inspection deters the ring only with enough capacity (with three inspected accounts per window in 5 of 6 markets in each family, with one in 2 of 6 and 3 of 6). Cancel fees and minimum resting times do not reduce spoofing, pump-and-dump or ring gains in the calibrated market and raise volatility by 6–24%. On real data, the 2016 tick-size reform of the Ho Chi Minh Stock Exchange lowered the share of zero-return days by 7–8 points relative to control exchanges, of the order of the simulator’s prediction for a halving of the tick and much smaller than for a five- or ten-fold cut; a market-adjusted abnormal-return rank places 8 of 10 documented manipulation cases in the daily top ten of about 1,360 stocks, against 15% for random stocks, and 701 pump events show an accumulation phase too weak for early warning. No real account-level data has been used.

1 Introduction

A securities regulator that wants to find manipulation faces three problems at once. It rarely has labelled cases, because enforcement is slow and selective. It cannot easily evaluate a rule change before introducing it, because the behaviour of manipulators changes with the rule. And the data that would let an outside researcher test a method, orders tagged with an account, is held only by the exchange and the regulator. These are acute in Vietnam. In ten recent decisions of the State Securities Commission with exact dates (Appendix A) the manipulation was carried out with between 5 and 164 accounts (median 24.5), by trading continuously, or crossing orders

among accounts, to create the appearance of supply and demand. Spoofing and layering, the focus of most of the detection literature [10, 30], do not appear among these cases.

We make four contributions. (i) We state a label-free coincidence test for coordinated accounts and prove that it controls false alarms exactly under a stated exchangeability condition (Section 4). (ii) We measure where it fails: against honest groups that also act in step, such as market-making desks, funds that split orders across sub-accounts and bot fleets reacting to one signal, and as the number of accounts grows (Section 5). (iii) We evaluate regulatory levers against a colluding ring that re-optimises its parameters, in a simulator calibrated to Vietnamese rules and to public price statistics (Section 6). (iv) We check what can be checked on real public data: documented enforcement cases against daily prices and a large set of labelled pump events on tick data (Section 7).

The paper does not claim that the detector works on real order data. Its manipulators and its features are written by us, a supervised model trained in the simulator does better than the label-free test in every scenario, and the real-data checks are at the level of the stock, not of the account. The contribution is a calibrated screening tool with a measured failure profile, and an evaluation method for rules that does not assume manipulators stand still.

2 Related work

Detection of manipulation. Machine-learning detectors for spoofing and layering work on the order book with labelled or simulated examples [10, 30]. Wash trading has been detected by graph rules on exchange data [31], by dynamic programming on matched orders [6], and by label-free statistical regularities of reported volume [8]. Collusion among traders has been approached through clustering of trade graphs [24] and through degree-strength statistics against randomised networks in Chinese trade data [29]; timing synchrony between traders has been described as a trait of performance rather than of collusion [28], and has been used to build networks of insiders from filings [14]. Pump-and-dump schemes organised on messaging channels have been studied with labelled events [12, 16, 17, 36] and priced as a market phenomenon [9]. We found no study that applies a time-shift coincidence test to the order submissions of accounts, and no study of account-level collusion with Vietnamese order data; our searches were not exhaustive.

Coincidence tests with surrogates. Testing whether events in two series coincide more than chance, without a model for the series, is a long-standing problem in neuroscience, where jitter and dither surrogates are used [2, 11, 20]. Interval jitter gives exact tests, whereas resampling that moves each event about its own position can inflate false positives [25]. Liang et al. [19] give finite-sample false-positive guarantees for time-shift tests that allow dependence within the series. Our statistic applies that idea to one account against the aggregate of all others, adds the netting of buys and sells, and takes the maximum over tolerances within a single shift distribution.

Simulation for rule design and for evaluating surveillance. Agent-based markets have been used to test tick size [38], price limits [37] and minimum resting times, cancellation fees, circuit breakers and transaction taxes [13], using open simulators [5]; their realism is a recognised issue [26, 33]. The closest work to ours on manipulation is the adversarial framework of Wang and Wellman [34], in which a generator learns to evade a discriminator, and the evaluation of a cloaking rule against an adaptive spoofer [35]. We differ by studying a ring rather than a single manipulator, by calibrating to a specific market, and by measuring regulatory levers by the best response of the manipulator. Our treatment of fines follows the economics of enforcement, in which deterrence depends on the product of the probability of detection and the sanction [4, 27];

double oracle and inspection games are the natural next step [3, 18, 21]. Empirical work on manipulation prevalence [1, 7, 15] and on Vietnamese markets [22, 23] motivates but does not overlap with our design.

3 A calibrated simulator

Market. A limit order book with price-time priority and self-trade prevention; one step is about one trading minute and a day has 240 steps. Honest agents are noise traders, market makers that re-quote two levels per side and shift with displayed depth imbalance, fundamental traders that follow a noisy latent value, capped momentum traders, depth-imbalance traders, and institutions that slice a parent order into market orders.

Manipulators. A *ring* of K accounts (3 to 40) runs cycles of accumulation by passive buys, a push by near-simultaneous aggressive buys from several members with some cross trades, and distribution by passive sells of the stock it holds; no member sells short. Orders are spread over members; a jitter parameter spreads the members’ actions in time. Outside its campaign each account trades as a noise trader. The ring’s benefit is a block held beforehand (3000 lots, an assumption) and sold off-book at the end of each push, valued at the achieved price displacement, plus trading profit. Single-account spoofers, pump-and-dump agents and wash pairs are also implemented.

Honest groups that act in step. A market-making firm with three to five desks that re-quote together; a fund that splits parent orders across 6 to 12 sub-accounts in the same step; and a fleet of bots that react to one public price signal with small latency differences.

Rules. The exchange can apply a tick size, minimum resting time, cancel fee, circuit breaker, daily price band, and a reference-price rule for off-book blocks (the last price or a trailing mean).

Calibration to Vietnam. Prices are in HOSE ticks: a stock at 25,000 VND with a 50 VND tick has a relative tick of 20 bp, and the daily band is $\pm 7\%$ of the previous close. Rules and statistics come from broker summaries and press (Appendix B); the exchange rulebooks could not be fetched. We add measurements from public datasets (Table 2). Table 1 compares the first preset with the real moments. Cancel share, relative tick and daily volatility are matched; the spread (3 ticks against one tick in 73% of futures quotes), the tails (excess kurtosis 4–5 daily and 40 at one minute) and volatility clustering are not.

Index level and stock level. The 1.16% target is the volatility of the VN30 *index*. A single stock is more volatile: the median HOSE stock has a daily volatility of 2.46% since 2022, and the stocks in the enforcement cases had a median of 1.75% in the year before the manipulation, with a share of zero-return days of 0.13 against 0.12 for HOSE stocks (Table 12). The case stocks are therefore ordinary HOSE stocks, not unusually illiquid ones, and the daily band was reached in their manipulation periods (median five days within 0.5 points of the ceiling; one stock for 12 days in a row). We therefore build two families of calibrated markets: index level (daily volatility 0.9–1.4%) and stock level (1.8–2.8%).

moment	simulator	real	source
cancel share of orders	0.321	0.318	HOSE, December 2020 (press)
relative tick (bp)	21	20	derived from the tick table
daily volatility, index level (%)	1.14	1.16	VN30, 2023 onward
daily volatility, typical stock (%)	–	2.46	HOSE median stock, 2022 onward
retail share of value	0.72	0.75–0.82	press estimates, 2024–25
spread: share of quotes at one tick	0	0.73	VN30F1M, 2024–25

Table 1: Calibration of the first Vietnam preset. The last row is not matched; the stock-level family targets the volatility of a typical stock.

data	content	used for
Daily prices of 1,551 Vietnamese stocks, 2015–2025	open, high, low, close, volume	stock-level detection checks (Section 7), tick-reform experiment (Section 6.4), case statistics
Exchange metadata	HOSE, HNX or UPCoM for each ticker	control group for the tick reform; band of each case stock
VN30, VN-Index, VN100 bars from 2012	daily and 30-minute bars	index-level volatility target
1.56 million VN30 futures ticks, 2024–25	price, best bid and ask	spread, tick activity (not matched)
10 documented enforcement cases	stock, exact period, accounts, decision	weak labels for detection checks
701 Telegram-organised pumps	tick trades around the pump, no accounts	accumulation before pumps, early-warning limits
Published rules and statistics	tick, band, lot, cancel share, retail share, fine cap	simulator rules and moments

Table 2: Real data used. None contains account identifiers.

4 A calibrated, label-free coincidence test

Take a window of T steps and n accounts. Let $x_i[t]$ be account i ’s number of buy submissions minus sell submissions at step t , and $y_{-i}[t] = \sum_{j \neq i} x_j[t]$. For a tolerance τ let $\tilde{y}_{-i}^\tau[t] = \sum_{|d| \leq \tau} y_{-i}[(t + d) \bmod T]$. For a shift $k \in \{0, \dots, T - 1\}$ define

$$C_i^\tau(k) = \sum_t x_i[(t - k) \bmod T] \tilde{y}_{-i}^\tau[t], \quad Z_i(k) = \max_{\tau \in \mathcal{T}} \frac{C_i^\tau(k) - \bar{C}_i^\tau}{s_i^\tau}, \quad (1)$$

where $\bar{C}_i^\tau$ and s_i^τ are the mean and standard deviation of $C_i^\tau(k)$ over k . The statistic is large when the account submits on the same side as the others at the same moments. The observed value is $Z_i(0)$ and the p -value is

$$p_i = \frac{1}{T} \#\{k : Z_i(k) \geq Z_i(0)\} \in [1/T, 1]. \quad (2)$$

All T shifts are computed by fast Fourier transform, so a window of n accounts costs $O(nT \log T)$ per tolerance. The default is $T = 300$ and $\mathcal{T} = \{1, 5, 15\}$. The unsigned variant keeps buys and sells as two channels and sums their statistics.

Proposition 1. *Suppose that under the null hypothesis the pair (x_i, y_{-i}) is invariant in law to a cyclic shift of x_i alone, that is, $(g x_i, y_{-i}) \stackrel{d}{=} (x_i, y_{-i})$ for every cyclic shift g . Then $\Pr(p_i \leq \alpha) \leq \alpha$ for every $\alpha \in [0, 1]$.*

Proof. Write $Z_k = Z_i(k)$ and $R_m = \#\{k : Z_k \geq Z_m\}$. Since $Z_k(x, y) = Z_0(g^k x, y)$, we have $R_m(x, y) = R_0(g^m x, y)$, and by invariance $R_0 \stackrel{d}{=} R_m$ for every m . For any real vector (Z_k) , at

detector	clean	+MM desks	+fund splitting orders	+bot fleet	all three
order-to-trade ratio	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.13 [0.05, 0.22]	0.09 [0.04, 0.14]
isolation forest (no labels)	0.01 [0.00, 0.02]	0.01 [0.00, 0.02]	0.00 [0.00, 0.00]	0.01 [0.00, 0.03]	0.02 [0.00, 0.03]
coincidence test, unsigned (no labels)	0.85 [0.78, 0.91]	0.32 [0.20, 0.46]	0.71 [0.65, 0.77]	0.86 [0.80, 0.91]	0.23 [0.15, 0.31]
coincidence test, signed (no labels)	0.87 [0.81, 0.92]	0.88 [0.82, 0.92]	0.74 [0.66, 0.80]	0.85 [0.79, 0.90]	0.52 [0.42, 0.63]
gradient boosting (labels, clean market)	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	0.99 [0.98, 1.00]	0.98 [0.96, 1.00]	0.98 [0.97, 1.00]

Table 3: catch@5 in a clean market and with honest groups acting in step (95% bootstrap intervals).

most $\lfloor \alpha T \rfloor$ indices m satisfy $R_m \leq \alpha T$. Hence $\Pr(R_0 \leq \alpha T) = \frac{1}{T} \sum_m \Pr(R_m \leq \alpha T) = \frac{1}{T} \mathbb{E} \#\{m : R_m \leq \alpha T\} \leq \alpha$. $\square$

The maximum over tolerances is taken inside $Z_i(k)$ for the same k , so it does not break the argument. The proposition is the standard randomisation-test argument; Liang et al. [19] treat the dependence within the series that makes it applicable to real time series. Its condition is strong. It fails when an account’s activity and the others’ activity share a trend inside the window, and when honest accounts respond to the same events; honest groups that act in step are precisely a violation, which is why we measure the false-alarm rate empirically (Section 5). Netting buys against sells before testing makes accounts that quote both sides at the same moment contribute nothing, while one-directional coordination does not cancel.

The test is used for screening. In each window the B accounts with the highest $Z_i(0)$ are the ones inspected, or accounts with $p_i \leq \alpha$ are flagged. A ring is *caught* if one member is inspected in some window, on the assumption that finding one member exposes the others, as account linking does in practice.

5 Detection experiments

All numbers in this section are for the default simulator (about 120 accounts in a window), one ring per episode, and episodes of 6000 steps (19 windows of 300 steps). Metric: *catch@B*, the share of ring-active windows in which one of the B top-ranked accounts is a ring member, with 95% bootstrap intervals over 24 test episodes.

Calibration. In a market without rings, 1.1% of honest accounts have $p \leq 0.01$ (market makers 0.7%, momentum traders 2.7%), in line with the nominal rate.

Robustness to honest synchronised groups. Table 3 and Figure 1 give catch@5 with and without honest groups. The three baselines that need no labels fail on the ring: the order-to-trade ratio is dominated by market makers, which sit far above manipulators, and an isolation forest on account features finds almost nothing. The unsigned test finds the ring in a clean market but collapses when market-making desks are present, because the desks, which quote both sides at once, are flagged at 71% (Table 4) and fill the top of the ranking. Netting buys against sells removes the effect of the desks (the desks’ flag rate falls to 0 in that scenario). A fund that splits orders across sub-accounts is genuinely coordinated and is flagged at 6% whatever the variant; it reduces catch@5 to 0.74. With all three groups together catch@5 is 0.52 for the signed test and 0.23 for the unsigned one. A gradient-boosting model trained with labels on rings in the clean market stays at 0.98–1.00 in every scenario.

Sensitivity. Figure 2 and Table 15 (Appendix) vary the design. The tolerance set and the window length matter little (catch@5 between 0.88 and 0.91). Sensitivity falls with the number

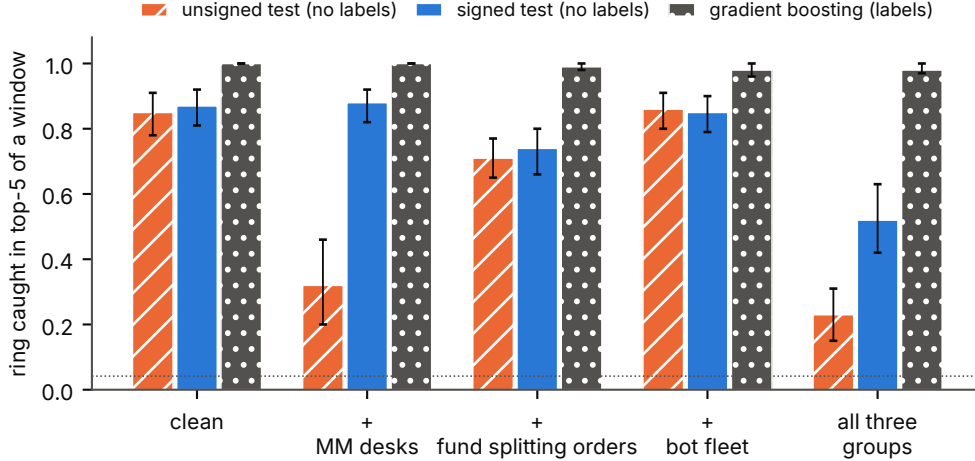


Figure 1: Share of ring-active windows in which a ring member is among the five top-ranked accounts, with 95% bootstrap intervals over 24 episodes. The dotted line is the chance level. The unsigned test is crowded out by market-making desks; the signed test is not. Gradient boosting is trained with labels on rings in the clean market.

scenario	honest group	members scored	flagged, unsigned	flagged, signed
+ MM desks	market-making desks	2622	0.711	0.000
+ fund splitting orders	fund sub-accounts	5277	0.061	0.063
+ bot fleet	bot fleet	3571	0.013	0.016
all three	market-making desks	2698	0.665	0.182
all three	fund sub-accounts	5202	0.050	0.053
all three	bot fleet	3732	0.029	0.037

Table 4: Honest group members flagged at $p \leq 0.01$ (nominal 1%).

of accounts: from 0.91 with 100 accounts to 0.84, 0.64 and 0.28 with 206, 373 and 708. It rises with ring size (0.79 for four accounts, 0.94 for 16 or more) and with the ring’s activity (0.46, 0.79, 0.91 for push probabilities 0.2, 0.4, 0.8). The dependence on the number of accounts is the main weakness of ranking by top- B ; a rule based on the p -value and a false-discovery-rate bound would scale differently, and we have not tested it.

What the label-free test is for. The model trained with labels is better in every scenario. The test is useful when there are no labels, which is the regulator’s position, and when one cannot trust that a model trained on simulated rings transfers to real ones. We cannot say which of the two would do better on real data.

6 Evaluating rules against manipulators that adapt

6.1 Design

Levers. The daily price band (7% today, 10%); the reference price of off-book block trades (the last price, or the mean over the last 60 or 240 steps); inspection of the B top-ranked accounts per window by the signed test, with a fine of m times the illegal gain; a minimum resting time before an order may be cancelled (5 or 15 steps); a fee per cancelled order; a larger tick; a circuit breaker. Vietnamese administrative fines are capped at five times the gain for individuals and

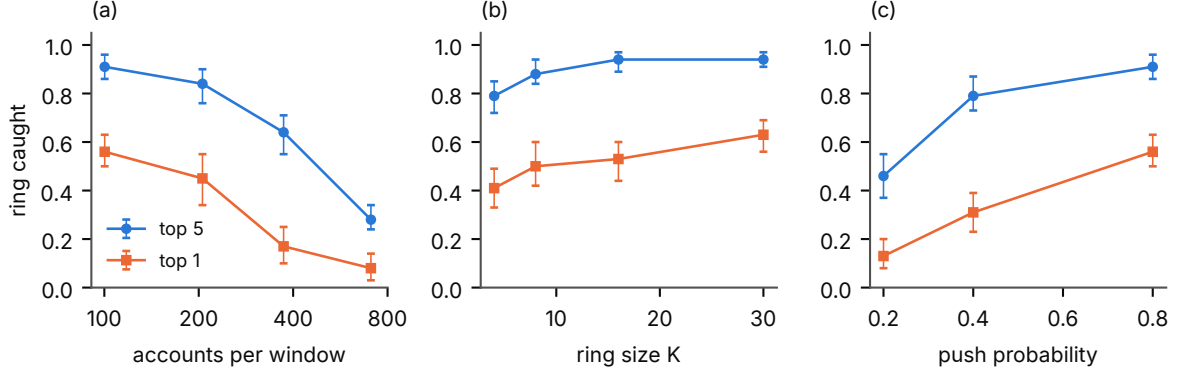


Figure 2: Sensitivity of the signed test in a clean market: (a) number of accounts per window, (b) ring size, (c) push probability of the ring. Bars are 95% bootstrap intervals over 16 episodes.

ten times for organisations (Appendix B), so we use $m = 5$ and $m = 10$. With a fine of m times the gain, deterrence needs a probability of detection above $1/m$ [4].

Manipulation types. The ring adapts: it chooses its parameters (size, order size, members per push, push and accumulation probabilities, jitter, cross-trade share, phase lengths, cool-down) to maximise its net benefit $U = g - m \max(g, 0) \mathbb{1}[\text{caught}]$. Because single evolution-strategy runs were noisy and non-monotone across policies, every policy faces the same set of ring strategies (the strongest found under each policy in one market, plus the default and the strongest unconstrained one); the ring picks the best on 12 selection episodes and the tables report 12 separate episodes. The result is a lower bound on a fully adaptive ring’s benefit. The spoofer, the pump-and-dump agent and the wash pair use fixed strategies and are evaluated for what they achieve; for them the effect of a rule is an upper bound.

A family of markets. A policy conclusion that holds in one calibration may not hold in a nearby one. We draw market parameters at random (numbers of noise and fundamental traders, market-maker activity and sensitivity to displayed depth, volatility of the latent value, momentum activity), simulate honest-only markets, and keep the draws whose moments lie within tolerance of the real ones (cancel share 0.27–0.37, retail share 0.66–0.85, and daily volatility 0.9–1.4% for the index-level family or 1.8–2.8% for the stock-level family), an approximate Bayesian rejection. We keep six markets per family and report the distribution of each outcome over markets.

6.2 The ring against each rule

Figure 3 and Table 5 give the ring’s gain in the index-level family. The daily price band does not change the ring’s gain (median ratio to no policy 0.97; the ring moves the price by about 0.5%, far inside a band of 7%), and widening it to 10% or 15%, which the securities commission is studying, changes nothing in the single market where we tried it (Appendix C). Pricing blocks at the trailing mean cuts the gain to a median 35% of its value at 60 steps and 11% at 240 steps (one day), and at 240 steps the gain is between -1k and 24k in every market against 87k to 162k with no policy. Inspection deters the ring only with enough capacity: with three inspected accounts per window (0.5% of accounts) the ring’s expected net benefit at $m = 5$ is non-positive in five of six markets, with one inspected account in two of six; the median probability of being caught is 42% and 17%.

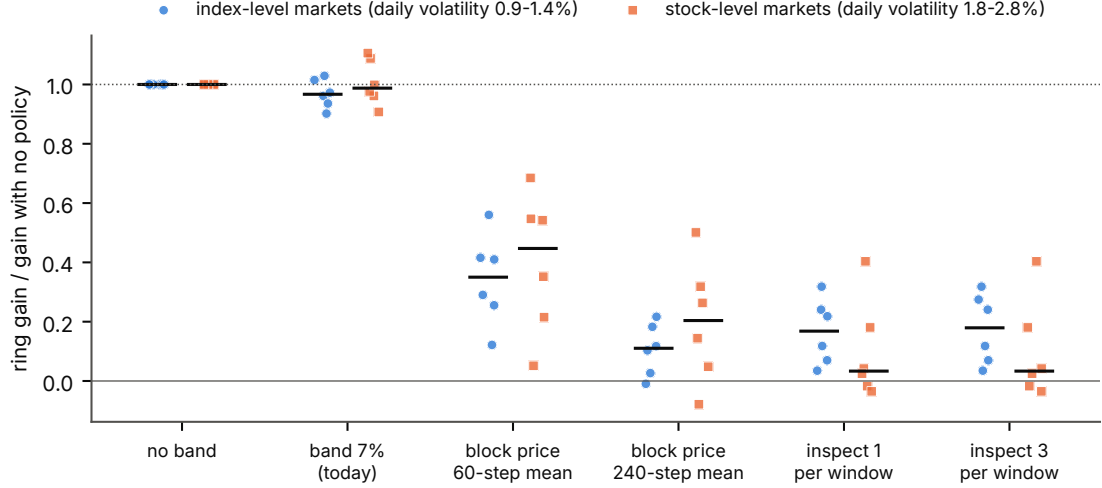


Figure 3: The best-responding ring’s gain relative to its gain with no policy, one point per calibrated market and the median as a line. Daily band 7% in all policies but the first.

policy	median gain	range over markets	median ratio to no policy	median detection prob.	deterred at $m = 5$
no price band	110.0k	86.7k to 162.1k	1.00	–	–
band 7% (HOSE today)	105.0k	89.2k to 164.6k	0.97	–	–
band 7% + block at 60-step mean	38.4k	11.7k to 90.8k	0.35	–	–
band 7% + block at 240-step mean	15.3k	-0.9k to 23.5k	0.11	–	–
band 7% + inspect 1 per window	16.8k	4.5k to 51.6k	0.17	17%	2/6
band 7% + inspect 3 per window	16.8k	4.5k to 51.6k	0.18	42%	5/6

Table 5: Index-level family of six calibrated markets: the best-responding ring’s gain (tick-lots) against each policy, the median probability that a member is inspected in some window of an episode, and the number of markets in which the ring’s expected net benefit is non-positive at a fine of five times the gain.

Stock-level family. The case stocks are more volatile than the index, so we repeat the exercise with six markets whose daily volatility is 1.8–2.8% (Table 6; same ring strategy set, not re-optimised). The ring’s gain with no policy is larger (median 164k against 110k at index level). The conclusions on the band and on block pricing are unchanged in sign and somewhat weaker in size: the band leaves the gain at a median 99% of its value, and block pricing cuts it to 45% (60 steps) and 20% (one day), but at one day the gain in one market is still 131k, so the rule does not remove the ring’s gain everywhere. Under inspection the best-responding ring shrinks its operation: its median gain falls to 3% of the no-policy gain (6k), with a median probability of being caught of 8% with one inspected account per window and 25% with three. The ring’s expected net benefit at $m = 5$ is non-positive in three of six markets with one inspected account and in five of six with three; in the remaining market with one inspected account the ring still nets up to 41k, and with three up to 7k. The detection probabilities are lower than at index level, where each window has fewer accounts to cover, yet deterrence is not worse, because the ring’s remaining gain is smaller relative to the fine; we have not separated the two effects. In both families the price band has no effect, block pricing and inspection have large ones, and no single rule removes the ring’s gain in every market.

6.3 Other manipulation types and the cost to honest investors

Table 7 reports, for the first calibrated market, what a fixed spoofer, pump-and-dump agent, wash pair and ring achieve under order-level rules and what the rules cost honest participants. In this market, minimum resting times and cancel fees do not reduce what the manipulators achieve:

policy	median gain	range over markets	median ratio to no policy	median detection prob.	deterred at $m = 5$
no price band	164.4k	101.2k to 261.4k	1.00	–	–
band 7% (HOSE today)	154.9k	111.9k to 261.1k	0.99	–	–
band 7% + block at 60-step mean	66.5k	8.4k to 179.0k	0.45	–	–
band 7% + block at 240-step mean	33.7k	-7.9k to 131.0k	0.20	–	–
band 7% + inspect 1 per window	6.3k	-9.1k to 40.8k	0.03	8%	3/6
band 7% + inspect 3 per window	6.3k	-9.1k to 40.8k	0.03	25%	5/6

Table 6: Stock-level family of six calibrated markets (daily volatility 1.8–2.8%): the ring’s gain against each policy, as in Table 5.

policy	spread	depth	volatility	retail cost	volume	spoof impact	pump impact	wash share	ring gain
band 7% (today)	+0%	+0%	+0%	+0%	+0%	0.11	0.47	0.033	34k
band 10%	-0%	-0%	+0%	-2%	-0%	0.14	0.43	0.033	32k
tick x2	+35%	-9%	+28%	-16%	+7%	0.21	0.40	0.031	30k
min rest 5	+2%	-1%	+6%	-5%	-0%	0.16	0.40	0.033	34k
min rest 15	+7%	-3%	+21%	-12%	+1%	0.26	0.40	0.032	33k
cancel fee 0.5	+2%	-2%	+6%	-5%	+1%	0.18	0.31	0.032	33k
cancel fee 2	+8%	-6%	+24%	-5%	+5%	0.20	0.31	0.031	36k
circuit breaker 6	-0%	-0%	-0%	+1%	-0%	0.15	0.45	0.033	33k
block at 60-step mean	+0%	+0%	+0%	+0%	+0%	0.11	0.47	0.033	20k

Table 7: First calibrated market, 10 honest-only and 24 manipulated episodes per policy. Spread, depth, volatility, retail trading cost and volume are relative to the band-7% market; spoofing and pump impact are in ticks; “wash share” is the share of volume traded between colluding accounts; ring gain is in tick-lots. Manipulators do not adapt. Retail cost includes fees paid; it falls under larger ticks and fees because simulated noise traders mostly post passive orders and earn the spread.

spoofing profit is 872 (s.e. 125) with no rule and 1,091 to 1,380 with minimum resting times or fees, and the spoofing price impact is higher, not lower, with a longer minimum resting time (0.26 against 0.11 ticks); pump impact is lower under cancel fees (0.31 against 0.47, within about 1.3 standard errors); wash volume is 3.3% of volume under every rule, and the ring’s gain is unaffected except by the block price. These rules cost honest investors: a minimum resting time of 15 steps raises the spread by 7% and volatility by 21%, and a cancel fee of 2 raises volatility by 24%. A tick twice as large widens the spread by 35% and cuts depth by 10% while lowering spoofing profit (491 against 872). The block price rule has no effect on honest trading in this experiment because honest agents do not trade blocks; its cost is the gap between the last price and the trailing mean, 0.17% of price at 60 steps and 0.50% at 240 steps in the first preset.

6.4 Checking a lever against reality: the 2016 tick-size reform

The tick size is the one lever in the simulator for which a real natural experiment is available in public data. On 12 September 2016 the Ho Chi Minh Stock Exchange reduced its minimum tick (to 10, 50 and 100 VND below 10,000, between 10,000 and 49,950, and above 50,000 VND), while HNX and UPCoM stocks were not affected [according to 32]. Using daily open, high, low, close and volume, we compare 282 HOSE stocks with 489 control stocks over the 40 trading days before and the 40 days from one week after the reform. Outcomes are the Corwin–Schultz high-low spread estimator, the daily range, the absolute return, the share of zero-return days and log volume. We could not establish the old tick from our sources (one summary gave 500 VND for all prices, which cannot be right given the small observed spread change), so we compare with simulated reductions of the tick by factors of 2, 5 and 10.

Table 8 shows the result. The share of zero-return days fell by 7.1 points relative to controls (95% interval -9.2 to -5.0), by 8.3 points with 20-day windows and by 6.7 with 60-day windows, a larger change than in any of four placebo dates; the spread estimate fell by 0.06 points (8% of its pre-reform level; interval -0.13 to 0.01), and by 0.07 points with 60-day windows, with

outcome	real DiD (95% interval)	real, relative to HOSE pre-period level	sim: tick 2 → 1	5 → 1	10 → 1
CS spread (%)	-0.061 [-0.133, 0.008]	-8%	-21%	-65%	-85%
daily range (%)	-0.067 [-0.213, 0.116]	-2%	-7%	-31%	-50%
return (%)	-0.141 [-0.270, 0.012]	-8%	-3%	-35%	-44%
zero-return share	-0.071 [-0.092, -0.050]	-24%	-38%	-43%	-78%
log volume	-0.000 [-0.153, 0.172]	—	—	—	—

Table 8: Tick-size reform: real difference-in-differences against the simulator’s prediction for a reduction of the tick by a factor of 2, 5 or 10. Changes relative to the pre-reform level.

lever	against rings	against spoofing, pumps, wash	cost to honest investors
inspection with a fine proportional to the gain	deters with enough capacity (three accounts per window: 5 of 6 markets in each family); depends on the detector	not tested	fine paid by manipulators only; inspection effort
off-book block priced at a trailing mean	cuts the ring’s gain to a median 35–45% (60 steps) or 11–20% (240 steps) of its value	not applicable	0.17%–0.50% of price on block trades
daily price band 7% or 10%	no effect in the markets tested	not tested	none observed in honest markets
minimum resting time, cancel fee	no effect	no reduction; spoofing profit and impact not lower	volatility +6% to +24%, spread up to +8%
tick size	not tested for rings	larger tick lowers spoofing profit	spread +35% for a doubled tick; real tick cut improved liquidity

Table 9: Policy evidence in this paper (simulated markets calibrated to Vietnam unless stated; the tick row also uses real data).

placebo estimates ranging from -0.06 to $+0.09$. By price tercile the fall in the spread estimate is concentrated in low-priced stocks (-0.18 points, interval -0.32 to -0.03 , against a pre-reform level of 1.0%). The simulator predicts a fall of the share of zero-return days by 38% for a halving of the tick, 43% for a five-fold cut and 78% for a ten-fold cut (real: 24%), and of the spread estimate by 21%, 65% and 85% (real: 8%). The signs agree; the real effect is of the order of the simulator’s prediction for a halving of the tick and much smaller than for a five- or ten-fold cut. Prices in the panel are adjusted, so price bands are approximate, and the design cannot separate the reform from other changes on HOSE in the same weeks.

6.5 Summary of the policy evidence

7 Checks on real public data

7.1 Documented enforcement cases on daily prices

Data and protocol. Ten decisions of the State Securities Commission with exact dates of the conduct were collected from press reports quoting the decisions (Appendix A; the decision documents were not opened). For every stock and date, using only data up to that date, we compute market-adjusted abnormal return, volume surge against the stock’s own history, a round-trip statistic, the position of the close in the daily range, the count of days with moves of at least 6.5%, and range expansion; stocks are ranked against each other on each date. The protocol, fixed before the cases were scored, is: a case stock is scored by the highest rank it reaches on any date in its period; two nulls are used, 300 random stocks over the same dates and 300 non-overlapping periods of the same length of the same stock; and we count the cases in

detector	superiority	top 10	chance	top 20	chance	top 50	chance
market-adjusted abnormal return	0.89	8/10	14%	10/10	29%	10/10	54%
combined six-feature score	0.79	6/10	22%	6/10	34%	10/10	57%
isolation forest, same features	0.79	7/10	20%	8/10	31%	9/10	53%
round trip (run-up, retracement)	0.70	9/10	75%	10/10	81%	10/10	90%
volume surge	0.33	0/10	3%	0/10	5%	0/10	13%

Table 10: Ten documented cases scored against about 1,360 stocks (20-day window). Superiority is the probability that a case stock beats a random stock (0.5 is chance); “top B ” counts cases in which the stock reached the daily top- B list at some date of its period; “chance” is the same share for random stocks.

window (days)	detector	superiority vs random stocks	superiority vs same stock	cases in top 10 / 10
10	abnormal return	0.87	0.78	9
10	combined score	0.69	0.70	5
10	volume surge	0.33	0.88	0
20	abnormal return	0.89	0.87	8
20	combined score	0.79	0.86	6
20	volume surge	0.33	0.90	0
60	abnormal return	0.85	0.89	3
60	combined score	0.77	0.88	3
60	volume surge	0.28	0.79	0

Table 11: Sensitivity to the window length. Superiority against random stocks and against the same stock at other times; number of cases in the daily top ten.

which the stock is among the B highest of its date at least once.

Results. Table 10 gives the results. The market-adjusted abnormal return alone places 8 of the 10 cases in the daily top ten of about 1,360 scored stocks at some point of their period, against 15% for random stocks over the same dates, and a case stock beats a random stock with probability 0.89 and the same stock at other times with probability 0.87. These are not stocks that rank high at all times. Combining six features, or an isolation forest on them, is no better than the single feature. Volume surge behaves differently in the two comparisons: against the same stock at other times it is informative (superiority 0.90, five of ten cases with $p < 0.05$), against other stocks it is not (0.33), because other stocks have larger surges at some time; we therefore claim nothing new for the market-level detector. Table 11 shows that the abnormal-return result is stable across window lengths of 10, 20 and 60 days (superiority over random stocks 0.87, 0.89, 0.85; over the same stock 0.78, 0.87, 0.89). The result says little about precision: periods are long (83 to 587 trading days), only ten cases are documented among thousands of price run-ups, and an unlabelled run-up is not an innocent one.

Case stocks and the daily band. Table 12 compares the case stocks with the market and counts days within 0.5 points of the daily ceiling. The stocks are ordinary (median daily volatility 1.75% in the year before against 2.46% for HOSE stocks), and the band is reached in their manipulation periods: five ceiling days at the median, and runs of up to 12 days. This differs from the simulated ring, whose price displacement of about 0.5% never reaches the band. The band could matter for a manipulator who ratchets the price over days; our simulated rings do not, so the finding that the band has no effect is a statement about rings of the size we simulate, not about all manipulation.

stock	exchange	days	pre-period vol. (%)	vol. in period (%)	zero-return share	log price change ($\times 100$)	ceiling days	longest run
FIR	HOSE	109	1.63	1.84	0.02	+38	0	0
SJS	HOSE	192	2.32	2.34	0.13	+49	10	3
PDR	HOSE	83	1.94	3.84	0.07	-119	4	3
PSH	unknown	319	1.75	3.45	0.45	-12	26	3
AGG	HOSE	514	1.29	1.90	0.13	+35	7	1
CRC	HOSE	106	1.36	2.65	0.18	+73	5	1
GKM	HNX	125	1.62	3.77	0.32	+157	1	1
PAS	UPCOM	491	3.37	3.54	0.10	-87	0	0
HCI	unknown	146	6.78	6.17	0.81	+183	21	12

Table 12: Case stocks: daily volatility and zero-return share in the year before the period, volatility during it, price change over the period, days within 0.5 points of the daily ceiling and the longest run of such days.

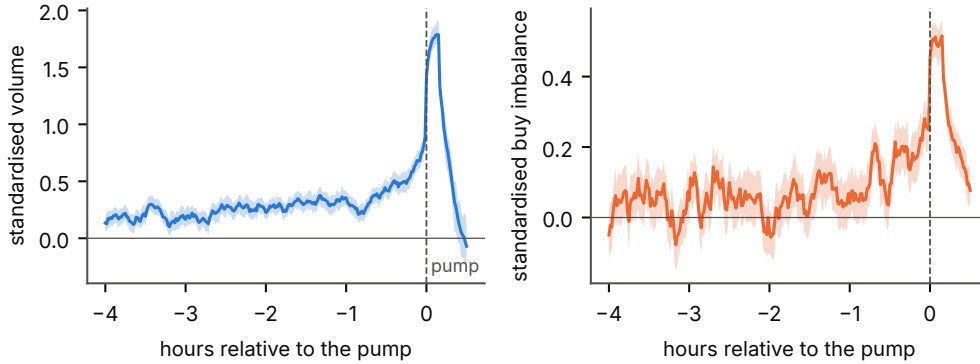


Figure 4: Standardised volume and buy imbalance of 701 pump events around the pump time (mean and 95% bootstrap interval over events).

7.2 Telegram-organised pump events on tick data

We use a public archive of 701 pump events with trades from about 24 hours before the pump time to shortly after (Appendix B). The files hold side, price and amount, with no account identifiers, so the coincidence test cannot be applied. For each event, standardised volume and buy imbalance over rolling ten-minute windows are measured against the same coin’s first 20 hours. Figure 4 shows accumulation: mean standardised volume rises from 0.19 (4 to 3 hours before) to 0.71 (last ten minutes), and buy imbalance from 0.03 to 0.23, with intervals excluding zero, and both jump at the pump. This supports the accumulate–push structure of the simulated ring. It does not give early warning: with the alarm rate in the background set to 0.05 per coin-hour, only 6.0% of events raise an alarm in the last hour before the pump (8.3% at 0.1 and 16.4% at 0.25 per coin-hour; median lead 7 to 15 minutes when it fires).

8 Discussion and limitations

What the evidence supports. For rings of the size simulated, in markets that match the real moments within tolerance: pricing off-book blocks at a trailing mean and inspecting accounts with a detector and a fine proportional to the gain remove most of the ring’s expected benefit, the second only with enough inspection capacity; the daily price band, minimum resting times and cancel fees do not, and the last two cost honest investors volatility. In real data, the tick reform improved liquidity in the direction the simulator predicts, and a simple abnormal-return rank flags documented cases far above chance, also against the same stock at other times.

What it does not support. No real account-level data has been used, so we do not know whether real rings are separable from honest coordination by this test; the simulator’s manipulators and honest groups were written by the same author as the detectors, and a model trained with labels in the simulator beats the label-free test everywhere. The calibrated markets do not reproduce the spread, the tails or volatility clustering. Rule facts come from broker summaries and press, and we could not establish the pre-reform tick. The ring search has a small budget and the ring’s strategy set was found in the first market; the strategies were not re-optimised in each market of the families, so the reported effects of the rules are lower bounds on what an adaptive ring gains. The block value is an assumption; a ring that is caught is assumed to be exposed by one inspected member; and fixed manipulators other than the ring are not best responses. The real-data checks are at the level of the stock, not of the account; the case periods come from press reports; the panel holds only stocks that are still listed; the natural experiment compares HOSE with other exchanges in the same weeks and cannot rule out other differences.

Next steps. Account-level data from an exchange, or per-account trade tables from case files, would allow a direct test; a rule based on p -values and a false-discovery-rate bound should replace top- B ranking, whose sensitivity falls with the number of accounts; the interplay of an adaptive ring with an adaptive regulator is a game that double-oracle methods could solve; and best responses should be computed in every market of the family.

Data and code availability

Code, simulator, experiment scripts, result tables and the case list are in the repository https://github.com/vinhq dang/quantitative_trading. The public datasets used are listed in Appendix B; raw files are not redistributed.

Ethics declaration

The study uses simulated data and public market data; no human subjects were involved. Enforcement cases are identified by stock ticker only; persons named in decisions are not named here.

Author contributions, funding, conflicts of interest

[To be completed by the author: CRediT contribution statement, funding statement, competing-interests statement.]

A Documented cases used in Section 7

B Sources for the calibration and real-data checks

- Tick sizes, daily bands, lot sizes, sessions: broker rule summaries (FPTS for HOSE, a securities-company PDF for HNX); the KRX system went live on 5 May 2025. Details and confidence tags: `docs/vietnam_calibration.md` in the repository.
- Fine cap of five times (individuals) and ten times (organisations) the illegal gain: Vietnamese press and legal-news summaries of Decree 156/2020 as amended; not read from the decree.

ticker	period	accounts	decided	type
FIR	2022-01-04 to 2022-06-17	76	SSC 2023-11 and 2024-01	cross trading among own accounts
SJS	2023-04-10 to 2024-01-10	26	SSC 2026-01-08	continuous buying and selling
PDR	2022-08-15 to 2022-12-09	164	SSC 2025-03-21	cross trading among own accounts
PSH	2021-02-01 to 2022-05-27	26	SSC 2024-05-27	negotiated trades within a group
AGG	2020-12-01 to 2022-12-31	92	SSC 2025-09 (year inferred)	continuous buying and selling
PPT	2022-06-17 to 2024-10-17	19	SSC decision 881	continuous buying and selling
CRC	2025-03-03 to 2025-08-01	10	SSC 2025-12-22	cross trading
GKM	2021-08-02 to 2022-01-28	23	SSC 2023-12-21	fake supply and demand to push the price up
PAS	2021-01-04 to 2022-12-30	19	SSC 2026-02-12	continuous buying and selling
HCI	2019-12-13 to 2020-07-16	5	SSC 2024-05-30	cross matched orders among own accounts
HNG	2015-07-20 to 2016-04-01	42	SSC 2017-08-10	multi-account fake supply and demand

Table 13: Cases with exact dates of the conduct. The periods are from press reports quoting the State Securities Commission decisions (sources are listed in the repository file `data/cases.csv`); the decision documents were not opened. FIR was the subject of two decisions with the same period and is counted once. PSH was traded on the negotiated system.

- Cancel and amendment share of 31.76% of orders (December 2020): press report of a HOSE statement.
- VN30, VN-Index and VN100 bars and the VN30F1M ticks: public Kaggle datasets (keithvo/vnstockdata, khimduong/vn30-market-making).
- Daily prices of 1,551 stocks: public Kaggle dataset (vuthinh/vietnam-stock-market-ohlc-price-data).
- Pump events: public Hugging Face dataset (Go3x3/pump_and_dump_dataset, MIT licence).

C Single-market results

The first version of the policy comparison used one calibrated market (index level, daily volatility 1.14%), with 20 selection and 20 reporting episodes per cell and seven strategies. Table 14 and Figure 5 are from it; they include the daily bands of 10% and 15% and a fine multiple of 10, and they are consistent with the family results (band: no effect; block price: large reduction; inspection: negative net benefit at one and three inspected accounts in that market).

policy	ring gain (mean, s.e.)	caught	net benefit after fine, $m = 5$	$m = 10$
no price band	81.7k (28.3k)	–	–	–
band 7% (HOSE today)	79.4k (29.3k)	–	–	–
band 10%	81.1k (28.2k)	–	–	–
band 15%	81.7k (28.3k)	–	–	–
band 7% + block at 60-step mean	21.7k (27.6k)	–	–	–
band 7% + block at 240-step mean	5.8k (21.4k)	–	–	–
band 7% + inspect 1 per window	13.8k (11.4k)	20%	-35.6k	-85.1k
band 7% + inspect 3 per window	13.8k (11.4k)	45%	-48.9k	-111.6k

Table 14: Single market: best-responding ring against each policy; net benefit with fines of $m = 5$ and $m = 10$ times the gain.

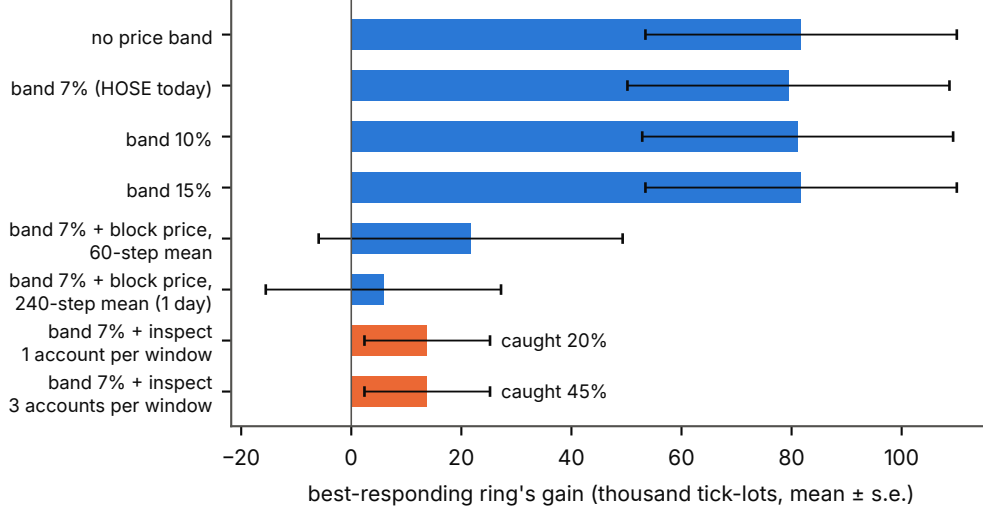


Figure 5: Single market: gain of the best-responding ring (thousand tick-lots, mean and standard error over 20 episodes).

D Additional tables

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factor	level	catch@5	catch@1	honest flagged	accounts per window
tolerances	(1,)	0.89 [0.85, 0.94]	0.44 [0.38, 0.52]	0.013	100
	(1, 5)	0.88 [0.83, 0.93]	0.55 [0.48, 0.62]	0.013	100
	(1, 5, 15)	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	(1, 5, 15, 30)	0.90 [0.86, 0.95]	0.55 [0.49, 0.61]	0.014	100
window length (steps)	150	0.88 [0.83, 0.91]	0.41 [0.34, 0.47]	0.007	81
	300	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	600	0.91 [0.84, 0.97]	0.58 [0.46, 0.71]	0.018	108
noise traders (accounts)	80	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100
	200	0.84 [0.76, 0.90]	0.45 [0.34, 0.55]	0.010	206
	400	0.64 [0.55, 0.71]	0.17 [0.10, 0.25]	0.009	373
	800	0.28 [0.24, 0.34]	0.08 [0.03, 0.14]	0.010	708
ring size K	4	0.79 [0.72, 0.85]	0.41 [0.33, 0.49]	0.013	93
	8	0.88 [0.84, 0.94]	0.50 [0.42, 0.60]	0.012	97
	16	0.94 [0.89, 0.97]	0.53 [0.44, 0.60]	0.012	104
	30	0.94 [0.91, 0.97]	0.63 [0.56, 0.69]	0.012	115
ring push probability	0.2	0.46 [0.37, 0.55]	0.13 [0.08, 0.20]	0.012	100
	0.4	0.79 [0.73, 0.87]	0.31 [0.23, 0.39]	0.014	100
	0.8	0.91 [0.86, 0.96]	0.56 [0.50, 0.63]	0.014	100

Table 15: Ablations and sensitivity of the signed test in a clean market (16 episodes per row, 95% bootstrap intervals). “Honest flagged” is the share of honest accounts with $p \leq 0.01$.

policy	spread	volatility	retail cost	spoofing impact	spoofing profit
baseline	2.6 ticks	0.21	2.2	1.07	56
tick $\times 4$	+124%	+50%	−54%	−25%	687
min. rest 5 steps	0%	−4%	+11%	+1%	13
min. rest 15 steps	+3%	+4%	+16%	−17%	−346
cancel fee 2	+1%	−1%	+36%	−5%	128
circuit breaker (6 ticks)	+1%	+2%	−6%	+3%	43

Table 16: Order-level rules against the default (non-adaptive) spoofer, 16 episodes per policy; changes relative to the baseline. Retail cost includes fees paid and falls under larger ticks because simulated noise traders mostly post passive orders and earn the wider spread.

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